VECTOR

MICROSAR Ethernet Transceiver Driver

Technical Reference

Marvell 88Q1010

Version 3.0.0

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Status	Released

Document Information

History

Author	Date	Version	Remarks
David Röder	2017-05-04	1.00.00	Creation of document
Martin Heiker	2018-11-26	2.00.00	STORYC-6938: Update TechRef to the newest core-version
Abhishek Halivan	2019-09-18	3.00.00	Hardware access monitoring

Reference Documents

No.	Source	Title	Version
[1]	Vector	TechnicalReference_EthTrcv_Core.pdf	see delivery

Scope of the Document

This complement to the technical reference [1] describes the derivative dependent aspects of the Ethernet Transceiver Driver basis software. Please refer to your Release Notes to get a detailed description of the platform (host, compiler, and transceiver) your Vector Ethernet Bundle has been configured for.



Caution
We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.



Note
This document describes aspects that apply for Ethernet Transceiver Driver modules provided by Vector, which are only used for Marvell 88Q1010 transceivers. For this reason, the corresponding infixes of APIs, macros, parameters etc. are replaced by 88Q1010.

The derivative independent aspects of the Ethernet Transceiver Driver are discussed in [1], where the infix mentioned above is replaced by the markers <Derivative> and <DERIVATIVE>.

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1 Component History

The component history gives an overview over the important milestones that are supported in the different versions of the component.

Component Version	New Features
01.00.xx	Initial Creation
02.00.xx	FEAT-3660: MSR4-R20 QM Release of DrvTrans_88Q1010EthAsr
03.00.xx	Hardware access monitoring introduced

Table 1-1 Component history

2 Introduction

This document describes the specific functionality of the driver either limiting or extending the function set described in [1].

In addition to that specific integration information for the driver itself and specific supported derivatives and use-cases beyond the general ones described in [1] are given.

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3 Hardware Overview

Table 3-1 summarizes the supported derivatives and settings this Ethernet Transceiver Driver can be used on. It also provides information on the documentation used during development of the driver for detailed reference to hardware specific information.

Derivatives	▶ 88Q1010
Compiler	Compiler independent
Hardware Manuals	▶ 88Q1010 Datasheet Automotive 100BASE-T1 Transceiver Doc. No. MV-S110544-00 Rev. A September 14, 2016, Draft
Safety Manuals	-
Errata Sheets	-
Specifics	-
Used Registers	See EthTrcv_30_88Q1010_LL_Regs.h
Hardware features related to independence or partitioning	-
Access Mechanism	▶ SMI management interface by utilizing Eth or EthSwt ▶ Ethernet switch internal registers by utilizing EthSwt
Hardware Diagnostics	▶ 4 PHY Test modes ▶ Internal/Remote Loopback mode ▶ Cable diagnostics ▶ Signal quality measurement

Table 3-1 Hardware Overview

4 Derivative Specific Functional Description

4.1 Overview

The general functional description of the transceiver driver is part of the general technical reference [1]. This section gives an overview over which of the features are supported by the specific transceiver driver considered in this document and to what extent. The features marked with ■ in Table 4-1 are available for the Ethernet transceiver driver for 88Q1010 transceivers. If a feature is supported with limitations, it is marked with ■. The limitations are explained in section 4.2.

Supported AUTOSAR Standard Conform Features	
Initialization of the Ethernet Transceiver	■
Setting and getting the Transceiver Mode	■
Transceiver wakeup	■
Restart auto negotiation	■
Get link state	■
Get baud rate	■
Get duplex mode	■
Set test mode	■
Set loopback mode	■
Get signal quality	■
Set TX mode	■
Get cable diagnostics result	■
Get PHY identifier	■
Retrieve version info	■
Report development errors to DET	■
Report production errors to DEM	■

Table 4-1 Supported AUTOSAR Standard Conform Features

4.2 Limitations to supported Features and Functions

4.2.1 PHY Test Modes

The PHY Test modes are explained in detail in the general technical reference [1]. In Table 4-2 all Test modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■ and unsupported marked with ■.

TestMode value	
ETHTRCV_PHYTESTMODE_NONE	■
ETHTRCV_PHYTESTMODE_1	■
ETHTRCV_PHYTESTMODE_2	■
ETHTRCV_PHYTESTMODE_3	■
ETHTRCV_PHYTESTMODE_4	■
ETHTRCV_PHYTESTMODE_5	■

Table 4-2 Supported PHY Test Modes

4.2.2 PHY Loopback Modes

The PHY Loopback modes are explained in detail in the general technical reference [1]. In Table 4-3 all Loopback modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■ and unsupported marked with ■.

LoopbackMode value	
ETHTRCV_PHYLOOPBACK_NONE	■
ETHTRCV_PHYLOOPBACK_INTERNAL	■
ETHTRCV_PHYLOOPBACK_EXTERNAL	■
ETHTRCV_PHYLOOPBACK_REMOTE	■

Table 4-3 Supported PHY Loopback Modes

4.2.3 PHY TX Modes

The TX modes are explained in detail in the general technical reference [1]. In Table 4-4 all TX modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■ and unsupported marked with ■.

TxMode value	
ETHTRCV_PHYTXMODE_NORMAL	■
ETHTRCV_PHYTXMODE_TX_OFF	■
ETHTRCV_PHYTXMODE_SCRAMBLER_OFF	■

Table 4-4 Supported PHY Tx Modes

5 Glossary and Abbreviations

5.1 Glossary

Term	Description
Platform	Hardware including host and communication controller (might also be integrated in host) on which the communication stack is implemented.

Table 5-1 Glossary

5.2 Abbreviations

Abbreviation	Description
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
BSW	Basis Software
Eth	Ethernet Controller Driver
EthTrcv	Ethernet Transceiver Driver
DEM	Diagnostic Event Manager
DET	Development Error Tracer
EAD	Embedded Architecture Designer
ECU	Electronic Control Unit
ISR	Interrupt Service Routine
MICROSAR	Microcontroller Open System Architecture (the Vector AUTOSAR solution)
MII	Media Independent Interface
PHY	Ethernet transceiver hardware, representing the interface to the physical layer
SMI	Serial Management Interface
SWS	Software Specification

Table 5-2 Abbreviations

6 Contact

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